Original Article

Diagnostic sensitivity of mammography and magnetic resonance imaging for ductal carcinoma *in situ* in high-risk breast cancer screening: A systematic-review and meta-analysis

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Abstract

Objective: Given the improved sensitivity of magnetic resonance imaging (MRI) for detecting ductal carcinoma *in situ* (DCIS), the omission of routine mammography (MG) or digital breast tomosynthesis (DBT) in high-risk breast cancer screening is under consideration. We aim to conduct a systematic review and meta-analysis to compare the screening sensitivity of MRI, MG and DBT for detecting DCIS in high-risk females.

Methods: PubMed, Embase, and Web of Science were searched for studies reporting the sensitivity of detecting DCIS in high-risk females up to July 02, 2025. Study quality was assessed with quality assessment of diagnostic accuracy studies-2 (QUADAS-2). Pooled sensitivity was estimated using a random-effects model, overall and stratified by age (<40 and ≥40 years old) and *BRCA* status (*BRCA1* and *BRCA2*). Meta-regression was used to compare modalities.

Results: Seventeen studies (18,348 participants, 211 with DCIS) were included. MRI showed significantly higher pooled sensitivity [85%, 95% confidence interval (95% CI): 74%−94%] than MG (36%, 95% CI: 23%−50%; P<0.001). No DBT data were available. Combined MRI and MG yielded the highest sensitivity (99%, 95% CI: 97%−100%), but offered no significant gain over MRI alone in females <40 years old (P=0.091) and in *BRCA1* mutation carriers (P=0.143).

Conclusions: MRI is more sensitive than MG for DCIS detection in high-risk females. In females <40 years old and *BRCA1* mutation carriers, adding MG to MRI provides no additional diagnostic value. Considering the potential trade-offs, the routine use of MG in these subgroups should be carefully reconsidered.

Keywords: Breast neoplasms; early detection of cancer; breast carcinoma *in situ*; mammography; magnetic resonance imaging; sensitivity and specificity

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Introduction

The average lifetime risk of developing breast cancer among females in the general Dutch population is approximately 13% (1 in 8) (1). However, females with an inherited predisposition to breast cancer are at higher risk. Among *BRCA1/2* mutation carriers, the cumulative risk to develop breast cancer by age 70 years is between 40% and 75% (2,3), often with an earlier onset compared to the general population (4). Females who received chest radiation before the age of 30 years also face elevated risk, with a cumulative breast cancer incidence of 13%−20% by age of 40−45 years (5). Other females at high risk include untested first-degree relatives of gene mutation carriers and

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those with multiple relatives diagnosed with early-onset breast or epithelial ovarian cancer (6).

High-risk females are advised to start annual breast magnetic resonance imaging (MRI) screening between 25 and 30 years old, supplemented with annual or biennial mammography (MG) starting ideally at the age of 35–40 years (7). MRI alone demonstrates high overall sensitivity for breast cancer detection, including both invasive cancers and ductal carcinoma *in situ* (DCIS) (75.2%–100%), with good specificity (83%–98.4%) (8). However, the types of tumors detected by MRI and MG differ: MRI is more likely to detect invasive cancers, whereas MG primarily contributes to the detection of DCIS (9), in particular through visualization of microcalcifications that are often not visible on MRI. Consequently, the added value of MG lies primarily in the detection of DCIS. For example, in the Dutch MRISC screening study, MG demonstrated a higher sensitivity for DCIS detection than MRI (69% *vs.* 38%) (10). Similarly, in the MARIBS study, only 2 of 6 DCIS cases were detected by MRI in contrast to 5 of 6 by MG (11).

However, with recent advances in MRI technology, MRI's ability to detect DCIS has markedly improved (12-14). Lo *et al.* found that among high-risk females, 12 out of 45 detected cancers (27%) were DCIS, all visible on MRI, with only one also detectable on MG (15). Riedl *et al.* observed a similar pattern, with all 14 DCIS cases detected by MRI and none by MG alone (13). Given the improved sensitivity of MRI for DCIS detection, and considering that adding MG reduces specificity by 0.4%–2% while increasing patient discomfort from compression, ionizing radiation exposure, and healthcare costs (8), the question of whether mammographic techniques [MG or digital breast tomosynthesis (DBT)] could be omitted from high-risk breast cancer screening has become a recent focus of study (15,16).

Since improving the detection sensitivity for DCIS is the main area where MG traditionally offers advantages, it is crucial to determine whether MRI alone achieves a detection sensitivity for DCIS comparable to that of combined imaging (MRI combined with MG or DBT), in order to evaluate whether MG can be omitted. To address this, we aim to conduct a systematic review and meta-analysis to compare the screening sensitivity of MRI, MG, and DBT for DCIS detection in females undergoing high-risk breast cancer screening.

Materials and methods

Our protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO; CRD42024622865) prior to data collection. This meta-analysis was performed in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis of Diagnostic Accuracy Studies (PRISMA-DTA) (17).

Literature search

A comprehensive literature search of PubMed, Embase and Web of Science was conducted to identify all relevant studies. The search was limited to English-language studies published between January 1, 2010 and July 02, 2025. The keywords used were: breast cancer, screening, DCIS, and sensitivity. References of retrieved studies were manually searched to identify additional studies for inclusion. The detailed search strategy is provided in the *Supplementary Table S1*.

Eligibility criteria

We included studies that met the following criteria: 1) the full text was available in English; 2) populations were females at high risk for breast cancer, defined as females with *BRCA* or other known genetic predispositions, untested first-degree relatives of a gene mutation carrier, females with history of chest radiation therapy between 10 and 30 years old and females with ≥20%–25% estimated lifetime risk based on risk assessment models that are largely dependent on family history (18); 3) females had undergone at least one imaging modality (MRI, MG or DBT); and 4) sufficient information was available to calculate the sensitivity of DCIS detection. The exclusion criteria are detailed in *Figure 1*. When multiple publications involved the same study population, we selected the one providing the most comprehensive data.

Data extraction and definitions

Two authors (M Li and Y Lin) independently performed data extraction using standardized data extraction forms. Any discrepancies were resolved by consensus or referral to a third reviewer (MDD). The following data were extracted: 1) study characteristics: first author, year of publication, journal of publication, country where the study was performed, study design, study setting; 2) study population: number of screened participants, number of DCIS cases, $\bar{x}\pm s$ or/and median (range) of patient age, risk criteria; 3) methods: screening modality, MRI parameters (MRI system, magnetic field strength and MRI sequence),

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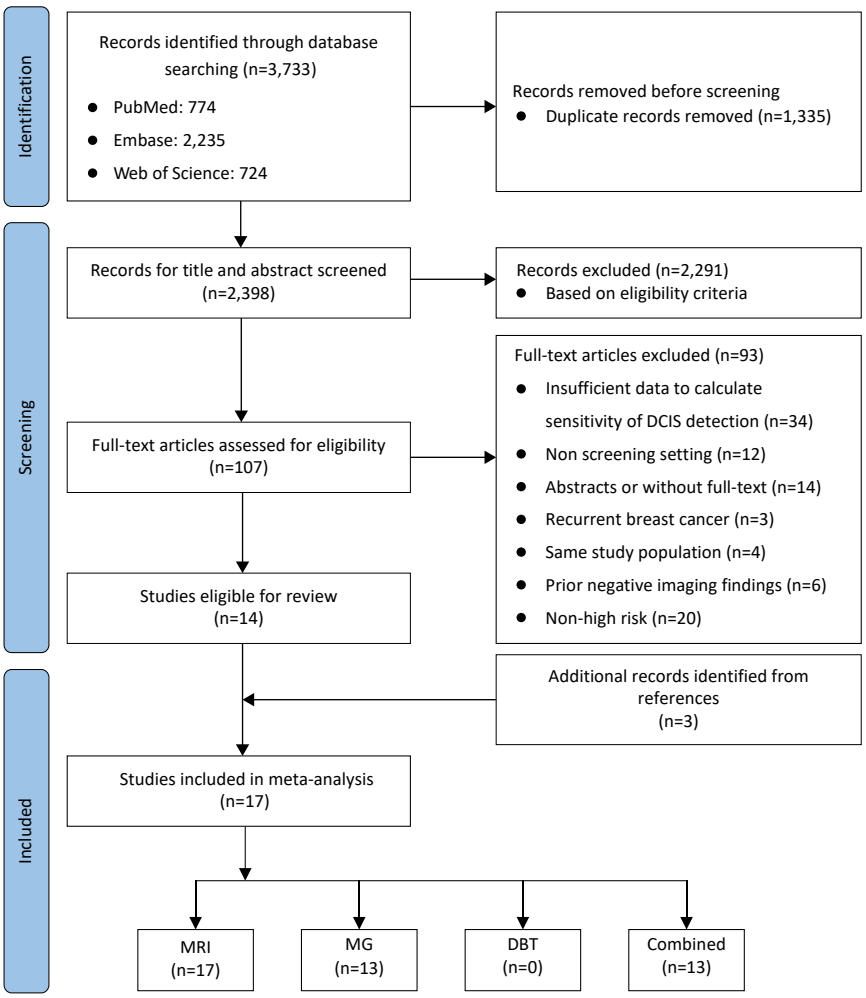


Figure 1 Flow chart of study selection. DCIS, ductal carcinoma *in situ*; MRI, magnetic resonance imaging; MG, mammography; DBT, digital breast tomosynthesis; Combined, combined MRI and MG.

MG type, reference standard used and threshold of Breast Imaging Reporting and Data System (BI-RADS) classification; and 4) analyzed diagnostic accuracy: true positives (TP) and false negatives (FN).

TP was defined as a case with a tissue diagnosis of DCIS following a positive screening result, and FN as a case with a tissue diagnosis of DCIS following a negative screening result.

Quality assessment

Two authors (M Li and Y Lin) independently assessed the methodological quality of the included studies using the quality assessment of diagnostic accuracy studies-2 (QUADAS-2) tool (19). A third investigator (MDD) was included in cases of disagreement. QUADAS-2 includes four domains: patient selection, index test, reference standard, and flow and timing. Each domain was evaluated for risk of bias, and the first three domains were also assessed for their relevance to the topic. The risk of bias for each domain was assessed using two or three signaling questions and subsequently judged as low, high, or unclear. Signaling questions for risk of bias are listed in *Supplementary Table S2*.

Statistical analysis

The outcome was the sensitivity of DCIS detection for MRI, MG, DBT alone, and MRI combined with MG or DBT. Sensitivity of DCIS was defined as the ability of the test to correctly identify DCIS in the screened population. Pooled estimates with 95% confidence interval (95% CI) were calculated based on a random-effects model. The Freeman-Tukey double arcsine transformation was applied

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to raw sensitivity data to normalize the distribution and stabilize variance (20), with results presented after back transformation. Forest plots were made to illustrate the pooled results and related heterogeneity. Heterogeneity among studies was assessed using Cochran Q test and I^2 index. For I^2 index: I^2<25% meant low heterogeneity; I^2 between 25% and 50% represented moderate heterogeneity; I^2>50% was considered substantial heterogeneity (21).

When substantial heterogeneity was observed, stratified and meta-regression analyses were performed to explore potential heterogeneity sources. Covariates examined included study region (North America *vs*. Europe), study design (prospective *vs.* retrospective), study setting (multi-center *vs.* single-center), BI-RADS threshold (≥3 or ≥4), and risk of bias (low *vs.* high). For MRI, type of MRI protocol (full-protocol *vs*. abbreviated breast MRI) and magnetic field strength (1.5 T *vs.* 3.0 T) were additionally analyzed. For MG, the type of modality (digital or film-screen or mixed) was also considered.

Comparative meta-regression analyses were conducted to compare the sensitivity of the following imaging modalities: 1) MRI *vs.* MG or DBT; and 2) MRI combined with MG or DBT *vs*. MRI alone, with the latter modality treated as the comparator group. For each meta-regression, beta coefficients with 95% CI, standard errors, and corresponding P values were reported.

Stratified analyses were further performed by age (<40 and ≥40 years) and *BRCA* status (*BRCA1* and *BRCA2*). In addition, to assess the robustness of the findings, sensitivity analyses were conducted by excluding studies published before 2015. In accordance with current guidelines for systematic review of diagnostic test accuracy, publication bias was not evaluated (17). Statistical analyses were performed using R software (Version 3.4.1; R Foundation for Statistical Computing, Vienna, Austria). P value less than 0.05 was considered statistically significant.

Results

Study selection

Initially, 3,733 records were identified through the literature search (*Figure 1*). After removing duplicates and reviewing titles and abstracts, 107 articles were retrieved for full-text review. Finally, 93 articles were excluded due to reasons such as insufficient data to calculate sensitivity of DCIS detection, non-high risk screening setting, abstracts or without full-text. This left 14 studies (10,12-16,22-29) for inclusion. An additional three studies (30-32) were included from the references, resulting in a total of 17 eligible studies. Reader agreement was substantial for both the title and abstract screening and the full-text screening (*Supplementary Table S3*).

Study characteristics

Of the 17 studies, all reported MRI sensitivity for DCIS detection (18,348 participants; 211 total DCIS cases; mean age: 43.3 years) (10,12-16,22-32), while 13 reported MG sensitivity (15,660 participants; 191 total DCIS cases; mean age: 42.7 years) (10,12-16,22-26,28,32). No studies addressed DBT sensitivity for DCIS. Thirteen studies reported combined MRI and MG sensitivity (15,660 participants; 191 total DCIS cases; mean age: 42.7 years) (10,12-16,22-26,28,32). Among the included studies, 11 were retrospective, and the remaining 6 were prospective. These studies were performed across North America (US 35%, 6/17; Canada 35%, 6/17) and Europe (the Netherlands 24%, 4/17; Germany 6%, 1/17). Seven of the 17 studies (41%) were published before 2015. *Supplementary Table S4* presents an overview of the characteristics of the included studies.

Technical details are reported in *Supplementary Table S5*. MRI scanners were predominantly from GE, Siemens, or Philips. Eight studies used 1.5 T (10,12,14,22,26,28,29,31), 4 studies used both 1.5 T and 3.0 T (15,23,24,30), 2 studies used 3.0 T (27,32), 1 study used 1.0 T and 1.5 T (13) and 2 did not report the information (16,25). Among the 17 studies, 16 used full-protocol MRI, all including dynamic contrast-enhanced (DCE) sequences with some variation: 5 combined DCE with T2-weighted (T2W) sequence (12-14,24,26), 6 included DCE, T2W, and T1-weighted (T1W) sequence (15,23,27,28,30,31), 3 reported DCE only (10,16,25), 1 employed DCE, T2W and diffusion-weighted imaging (32), and one study did not report the MRI protocol (22). Only 1 study used an abbreviated breast MRI protocol, consisting of precontrast and first postcontrast T1W images alone with a T2W sequence (29). For MG type, eight studies used digital MG (15,16,22-25,28,32), while the remaining five used both digital and screen-film MG (10,12-14,26).

Quality assessment

Fourteen studies of 17 studies were at low risk of bias (12-16,22-27,29,30,32). The most common sources

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contributing to a high risk of bias were related to flow and timing, primarily due to the exclusion of some screened participants from the evaluation of the imaging modalities' performance (10,28). The second most common risk factor for bias was observed in index test, where imaging findings interpreted according to BI-RADS criteria had unclear thresholds (31). All studies were rated low risk of bias for patient selection and reference standard, and no applicability concerns were identified in any of QUADAS-2 domains (*Supplementary Figure S1*, *Table S6*).

Pooled analysis

Forest plots and pooled sensitivity estimates of MRI, MG and combined MRI and MG overall and stratified by age and *BRCA* status are shown in *Figures 2–4*, *Supplementary Figures S2–S7* and summarized in *Table 1*. A summary of the meta-regression results comparing MRI *vs*. MG and combined MRI and MG *vs*. MRI alone is shown in *Table 2*. MRI demonstrated a pooled sensitivity of 85% (95% CI: 74%–94%), which was significantly higher than that of MG at 36% (95% CI: 23%–50%; P<0.001). Combined MRI and MG yielded the highest pooled sensitivity at 99% (95% CI: 97%–100%), exceeding that of MRI alone (P=0.001) (*Tables 1*,*2* and *Figures 2–4*).

Heterogeneity

Quantification of heterogeneity using the I^2 statistic is presented in *Table 1*. Substantial heterogeneity was observed for both MRI (I^2=54%) and MG (I^2=60%), whereas combined MRI and MG showed no substantial heterogeneity (I^2=0). Stratified and meta-regression analyses were performed to further explore the sources of heterogeneity (*Supplementary Table S7*). For MRI, risk of bias was identified as a significant contributor, with low-risk studies showing higher sensitivity than high-risk studies (89% *vs.* 66%, P=0.050). For MG, heterogeneity was primarily explained by study region and study setting, with higher sensitivity in Europe *vs.* North America (58% *vs.* 23%, P<0.001) and in multi-center compared to single-center studies (45% *vs*. 19%, P=0.043).

Stratified analysis by age and BRCA status

In the age-stratified analysis, the sensitivity of MRI, MG, and combined MRI and MG was 83% (95% CI:

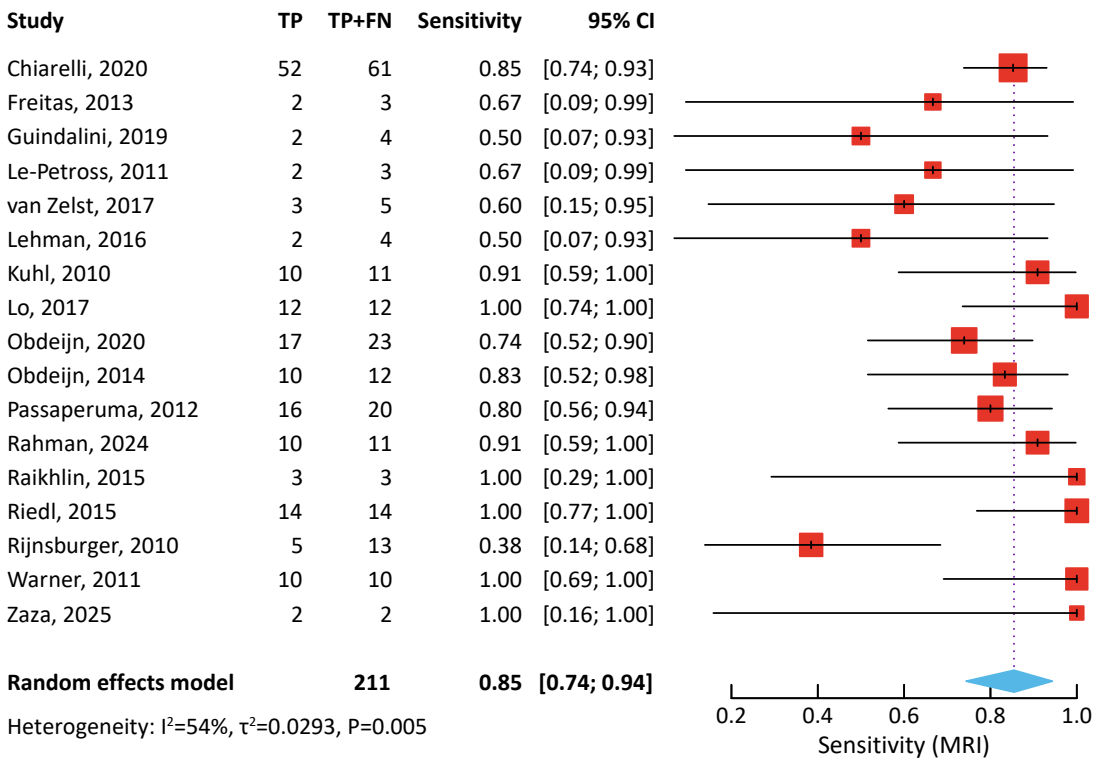


Figure 2 Pooled sensitivity forest plots for MRI to detect DCIS. MRI, magnetic resonance imaging; DCIS, ductal carcinoma *in situ*; TP, true positive; FN, false negative; 95% CI, 95% confidence interval.

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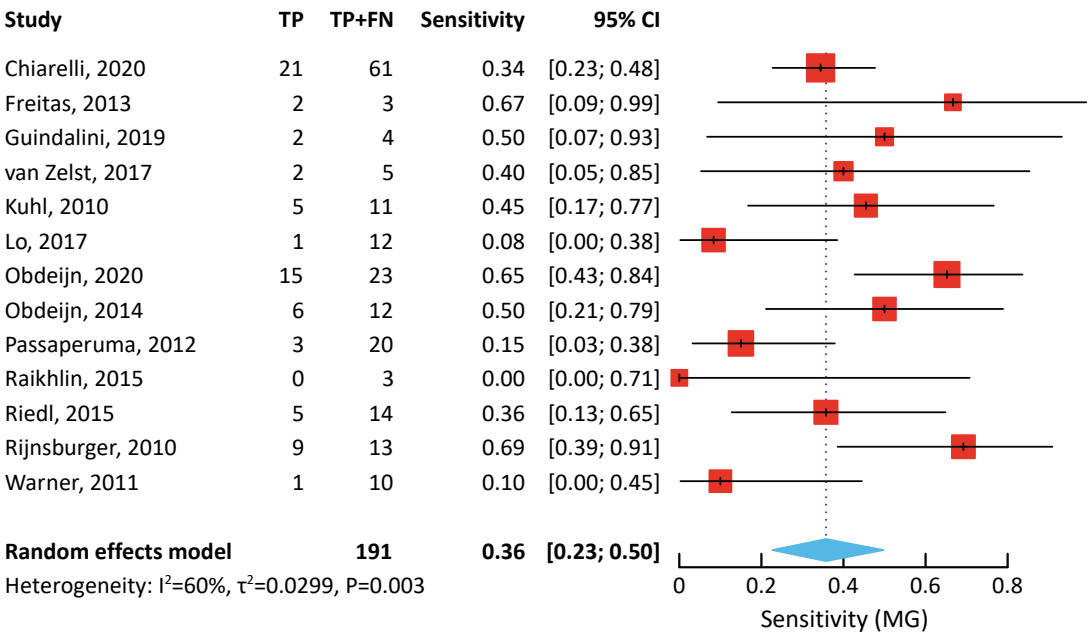


Figure 3 Pooled sensitivity forest plots for MG to detect DCIS. MG, mammography; DCIS, ductal carcinoma *in situ*; TP, true positive; FN, false negative; 95% CI, 95% confidence interval.

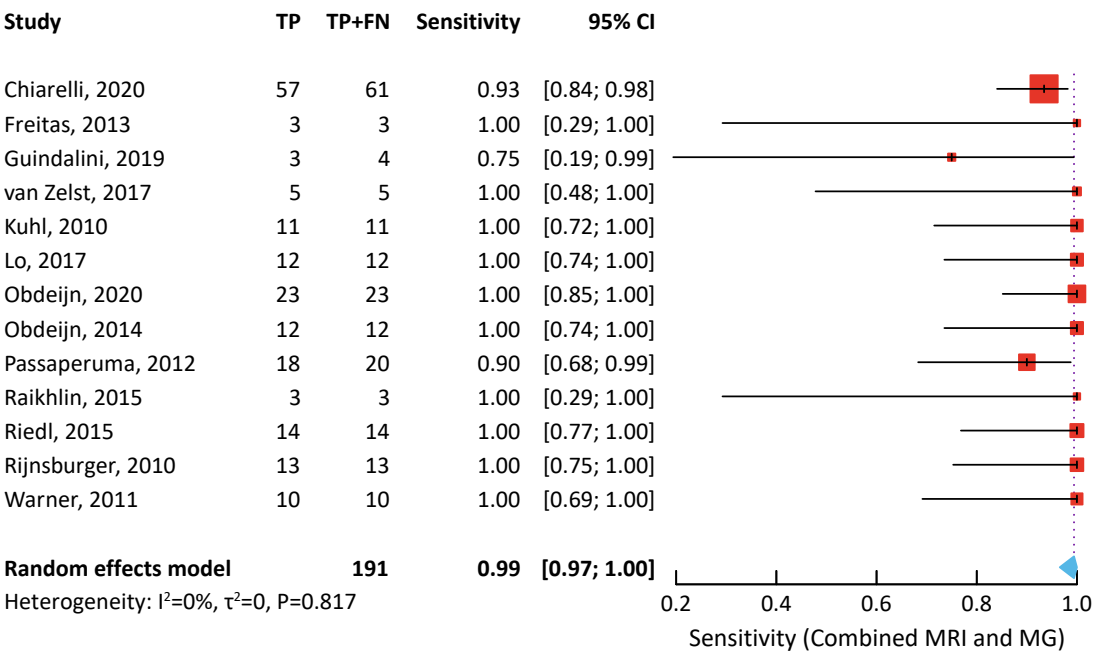


Figure 4 Pooled sensitivity forest plots for combined MRI and MG to detect DCIS. MRI, magnetic resonance imaging; MG, mammography; DCIS, ductal carcinoma *in situ*; TP, true positive; FN, false negative; 95% CI, 95% confidence interval.

60%–99%), 27% (95% CI: 6%–54%), and 100% (95% CI: 87%–100%) among females <40 years, respectively. The difference between MRI and MG was statistically significant (P<0.001), whereas combined MRI and MG did not significantly improve sensitivity compared with MRI alone (P=0.091). Among females ≥40 years, sensitivities were 90% (95% CI: 73%–100%), 28% (95% CI: 9%–50%), and 100% (95% CI: 99%–100%), respectively. In this group, MRI was significantly more sensitive than MG (P<0.001), and combined MRI and MG achieved a

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significantly higher sensitivity than MRI alone (P=0.012) (*Tables 1*,*2*, *Supplementary Figures S2–S4*).

In the *BRCA* status-stratified analysis, MRI, MG, and combined MRI and MG showed sensitivities of 95% (95% CI: 79%–100%), 15% (95% CI: 0–46%), and 100% (95% CI: 96%–100%) among *BRCA1* mutation carriers, respectively. MRI was significantly more sensitive than MG (P<0.001), while the difference between combined MRI and MG and MRI alone was not significant (P=0.143). For *BRCA2* mutation carriers, the corresponding sensitivities were 84% (95% CI: 67%–97%), 33% (95% CI: 10%–60%), and 100% (95% CI: 96%–100%). In this group, MRI sensitivity was significantly higher than that of MG (P=0.005), and combined MRI and MG demonstrated significantly greater sensitivity than MRI alone (P=0.001) (*Tables 1*,*2*, *Supplementary Figures S5–S7*).

Sensitivity analysis

After excluding seven studies published before 2015, the pooled sensitivities for MRI, MG, and combined MRI with MG were 89% (95% CI: 76%–98%), 34% (95% CI: 17%–53%), and 99% (95% CI: 96%–100%), respectively. MRI remained consistently more sensitive than MG, except among *BRCA2* mutation carriers. Combined MRI and MG did not significantly improve sensitivity in females under 40 years old or in *BRCA1* mutation carriers compared to MRI alone (*Supplementary Tables S8*,*S9*). While the results showed slight variations, they remained largely consistent with the primary analysis.

Discussion

This systematic review and meta-analysis comprehensively compared the sensitivity of MRI, MG and combined MRI and MG for DCIS detection in high-risk breast cancer screening. Our findings indicated that MRI had a significantly higher sensitivity than MG (85% *vs.* 36%), while no data were available on DBT. Combined MRI and MG demonstrated the highest sensitivity at 99%. Stratified analysis revealed that MRI consistently showed higher sensitivity than MG across all subgroups. However,

Table 1 Sensitivity (%) and 95% CI for DCIS detection overall and stratified by age and *BRCA* status

Stratification	MRI			MG			Combined		
	n†	Sensitivity (95% CI)	I^2 (%)	n†	Sensitivity (95% CI)	I^2 (%)	n†	Sensitivity (95% CI)	I^2 (%)
Overall	17	85 (74–94)	54	13	36 (23–50)	60	13	99 (97–100)	0
Age at diagnosis (year)									
<40	11	83 (60–99)	0	10	27 (6–54)	7	9	100 (87–100)	0
≥40	11	90 (73–100)	44	9	28 (9–50)	37	8	100 (99–100)	0
BRCA status									
BRCA1	9	95 (79–100)	0	8	15 (0–46)	42	7	100 (96–100)	0
BRCA2	8	84 (67–97)	20	7	33 (10–60)	41	6	100 (96–100)	28

DCIS, ductal carcinoma *in situ*; MRI, magnetic resonance imaging; MG, mammography; Combined, combined MRI and MG; 95% CI, 95% confidence interval. †, No. of studies.

Table 2 Comparative meta-regression analysis of sensitivity for DCIS detection overall and stratified by age and *BRCA* status

Stratification	MRI *vs.* MG			Combined *vs.* MRI		
	β coefficients (95% CI)	Standard error	P	β coefficients (95% CI)	Standard error	P
Overall	0.46 (0.29–0.63)	0.09	<0.001	0.24 (0.11–0.37)	0.07	0.001
Age at diagnosis (year)						
<40	0.39 (0.15–0.64)	0.12	<0.001	0.22 (–0.04–0.48)	0.13	0.091
≥40	0.53 (0.29–0.78)	0.12	<0.001	0.24 (0.05–0.43)	0.10	0.012
BRCA status						
BRCA1	0.64 (0.37–0.91)	0.14	<0.001	0.18 (–0.06–0.43)	0.13	0.143
BRCA2	0.37 (0.11–0.62)	0.13	0.005	0.34 (0.13–0.54)	0.11	0.001

DCIS, ductal carcinoma *in situ*; MRI, magnetic resonance imaging; MG, mammography; Combined, combined MRI and MG; 95% CI, 95% confidence interval.

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combined MRI and MG provided no substantial increase in sensitivity compared to MRI alone in females younger than 40 years and *BRCA1* mutation carriers.

While the superior diagnostic sensitivity of MRI compared to MG for invasive breast cancer is well established (33), its value in detecting DCIS remains controversial. In some cases, MG can depict MRI-occult DCIS by identifying microcalcifications (34,35). Consequently, routine screening MG continues to be recommended for all females at high risk. However, in our study, we found that MG offers limited sensitivity of 27% and the addition of MG to MRI provided no significant added benefit among females younger than 40 years. This limited contribution of mammography in young populations has also been noted in previous studies (22,25,36). One possible explanation is the high density of glandular tissue in young females. Dense tissue appears bright on a mammogram (37), which can obscure subtle signs of DCIS, thereby reducing MG sensitivity. Additionally, the low prevalence of calcified DCIS lesions in this age group may further contribute to the limited effectiveness of MG (38).

MG has differential value between *BRCA1* and *BRCA2* mutation carriers. Our results showed that in *BRCA1* carriers, MG demonstrated lower sensitivity compared to *BRCA2* carriers (15% *vs.* 33%), and the addition of MG to MRI provided no significant benefit. Conversely, for *BRCA2* carriers, MG significantly improved DCIS detection when combined with MRI. A similar observation was reported in a meta-analysis conducted by Phi *et al.* (39), which concluded that the additional screening sensitivity provided by MG is limited in *BRCA1* mutation carriers, whereas MG meaningfully contributes to screening sensitivity in *BRCA2* mutation carriers. This difference may be attribute to variation in tumor histology (40), as DCIS is more frequently observed in *BRCA2*-associated breast cancer and sometimes only visible as mammographic calcifications (35).

Our findings indicate that MG offers limited added value in addition to MRI, further supporting evidence for the potential omission of MG in high-risk breast cancer screening. A meta-analysis by Ding *et al.* examining screening outcomes in females at high risk found that MRI alone increased the breast cancer detection rate by 8 per 1,000 females compared MG alone, while adding MG to MRI provided only marginal improvements (41). This finding is consistent with the study conducted by Lo *et al.*, which demonstrated MG did not detect any additional cancers beyond those identified by MRI in high-risk females (15). Furthermore, studies by Kuhl *et al.* (12) and Riedl *et al.* (13), which focused on females with a high familial risk, also concluded that MG does not provide substantial added value in the detection of breast cancer when MRI is used. For *BRCA*1 mutation carriers, particularly those under the age of 40 years, studies consistently show that they do not benefit from adding MG to MRI screening (16,22,42).

This study has some limitations. First, pooling specificity data for DCIS detection and generating receiver operating characteristic curves was not feasible, as screening studies report specificity for overall breast cancer (invasive cancer + DCIS). Separate calculation of specificity for invasive and DCIS is not possible because imaging cannot reliably distinguish between the two before biopsy. However, given the high sensitivity of MRI for invasive cancer and its potential to improve specificity by avoiding MG-related false positives, comparing DCIS sensitivity across modalities sufficiently captures the information needed to address our research question on whether MG could be omitted in high-risk screening. Second, no data were available on the sensitivity of DBT for DCIS detection in high-risk screening, which limit our ability to assess its diagnostic sensitivity compared to MRI and MG. This is probably because DBT is not routinely used in females at high risk (7). Although a growing body of research exists on DBT, most studies focused on average-risk populations (43). Some evidence is available in females at intermediate risk, including those with dense breasts (44) or a personal or family history of breast cancer (45,46). Third, 41% of the included studies were published over a decade ago; however, sensitivity analyses excluding these studies demonstrated consistent results, supporting the robustness of our conclusions. Fourth, the lack of studies from population outside North America and Europe restricts the generalizability of our findings, as difference in genetic predisposition, breast density, and healthcare practices may influence screening outcomes. Fifth, artificial intelligence tools like Rayyan can facilitate literature screening, but their reliability depends on manual verification. In our study, all potential duplicates flagged by Rayyan were carefully checked and confirmed manually, which minimized the risk of erroneous exclusion. Finally, we exclude non-English language studies, which may have introduced publication bias and may have led to the omission of relevant studies.

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Conclusions

Our meta-analysis demonstrates that MRI is significantly more sensitive than MG for DCIS detection in high-risk females. However, in females younger than 40 years and *BRCA1* mutation carrier, adding MG to MRI does not significantly improve sensitivity. Given that MG does not significantly improve sensitivity when MRI is performed, and considering its associated false-positive rates, radiation exposure and costs, the use of combined MRI and MG in a screening setting must be applied with caution. Our findings support the use of MRI as the primary screening modality, where MG can potentially be omitted in females younger than 40 years and *BRCA1* mutation carriers.

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Author contributions

Conceptualization, data curation, formal analysis, methodology, writing − original draft, writing − review & editing: MM Li; Data curation, writing − review & editing: YX Lin; Conceptualization, writing − review & editing: K Poelhekken; Conceptualization, methodology, supervision, writing − original draft, writing − review & editing: MJW Greuter, de Bock GH and Dorrius MD.

Footnote

Conflicts of Interest: The authors have no conflicts of interest to declare.

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